

Q1: Aspect Term Extraction: Model Performance Analysis

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Abstract

This report presents the design, implementation, and evaluation of deep learning models for Aspect Term Extraction (ATE), a subtask of Aspect-Based Sentiment Analysis. Four different neural models were implemented and compared: RNN with GloVe embeddings, RNN with FastText embeddings, GRU with GloVe embeddings, and GRU with FastText embeddings. The models were trained on a dataset of annotated restaurant reviews and evaluated using F1-score. The GRU model with FastText embeddings achieved the best performance with an F1 score of 0.6587. This report details the preprocessing steps, model architectures, training process, and comparative analysis of the models.

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1 Introduction

Aspect-Based Sentiment Analysis (ABSA) is a fine-grained sentiment analysis task that aims to identify the specific aspects of entities mentioned in text and the sentiment expressed towards them. Aspect Term Extraction (ATE) is a crucial subtask of ABSA that involves identifying explicit mentions of aspects or opinion targets in text. For example, in the restaurant review sentence "The food was delicious but the service was slow," the aspect terms are "food" and "service."

This report details the implementation and evaluation of several neural network models for the ATE task. The models are based on recurrent neural network architectures (RNN and GRU) and utilize pre-trained word embeddings (GloVe and FastText). The models are trained and evaluated on a dataset of annotated restaurant reviews.

The main contributions of this work include:

- Implementation of four neural network models for ATE
- Comprehensive comparison of model architectures and embedding types
- Detailed analysis of model performance and error patterns
- Development of a testing function for model evaluation on unseen data

2 Task Description

Aspect Term Extraction involves identifying the specific aspects of products or services that customers are referring to in their reviews. In the restaurant domain, these aspects might include the food, service, ambience, price, and so on.

The task is formulated as a sequence labeling problem, where each word in a sentence is assigned one of the following tags:

- B-TERM: Beginning of an aspect term
- I-TERM: Inside (continuation) of an aspect term
- O: Outside (not part of any aspect term)

This BIO (Beginning-Inside-Outside) tagging scheme is a standard approach for sequence labeling tasks in natural language processing. The models must learn to identify aspect terms from context without explicit syntactic or semantic guidance, making this a challenging task.

3 Dataset

The dataset consists of annotated restaurant reviews. Each example in the dataset contains a sentence, aspect terms with their polarities, and aspect categories. Here is an example from the dataset:

```
1 {
2   "sentenceid": "3534",
3   "sentence": "All the money went into the interior
4     decoration, none of it went to the chefs.",
5   "aspect_terms": [
6     {
7       "term": "interior decoration",
8       "polarity": "positive",
9       "from": "28",
10      "to": "47"
11    },
12    {
13      "term": "chefs",
14      "polarity": "negative",
15      "from": "72",
16      "to": "77"
17    }
18  ],
19  "aspect_categories": [
20    {
21      "category": "ambience",
22      "polarity": "positive"
23    },
24    {
25      "category": "food",
26      "polarity": "negative"
27    }
28  ]
29 }
```

For each aspect term, the dataset provides:

- The exact term text
- The sentiment polarity (positive, negative, or neutral)
- Character indices indicating the span of the term in the sentence

The dataset is split into training and validation sets, stored in `train.json` and `val.json` respectively.

4 Preprocessing

4.1 BIO Encoding

The first step in preprocessing was to convert the aspect annotations into the BIO tagging scheme. This involved:

1. Tokenizing the sentences using a simple regex pattern (`\b\w+\b|[\^\w\s]`) to capture both words and punctuation
2. Initializing all token labels as 'O' (Outside)
3. Finding the tokens that correspond to aspect terms and updating their labels:
 - The first token of an aspect term is labeled as 'B-TERM'
 - Subsequent tokens of the same aspect term are labeled as 'I-TERM'

The preprocessing function extracts exact aspect terms from the sentence using the character indices provided (`from` and `to`), tokenizes these terms, and then matches them against the tokenized sentence to assign proper BIO tags.

```
1 def preprocess_data(file_path, output_path):
2     logger.info(f"Preprocessing {file_path}...")
3     preprocessed_data = []
4
5     with open(file_path, 'r', encoding='utf-8') as f:
6         data = json.load(f)
7
8     for item in data:
9         sentence = item['sentence']
10        aspect_terms = item.get('aspect_terms', [])
11
12        # Simple tokenization
13        tokens = re.findall(r'\b\w+\b|[\^\w\s]', sentence)
14
15        # Initialize all labels as 'O'
16        labels = ['O'] * len(tokens)
17
18        # Extract aspect terms
19        aspect_terms_list = []
20
21        for aspect in aspect_terms:
22            term = aspect['term']
23            from_idx = aspect['from']
24            to_idx = aspect['to']
```

```

26         # Extract the exact term from the sentence
27         aspect_term = sentence[int(from_idx):int(
to_idx)]
28         aspect_terms_list.append(aspect_term)
29
30         # Find the tokens that correspond to this
aspect term
31         aspect_tokens = re.findall(r'\b\w+\b|[\^\w\s]',
, aspect_term)
32
33         # Look for these tokens in the original
tokens list
34         for i in range(len(tokens) - len(
aspect_tokens) + 1):
35             if [token.lower() for token in tokens[i:i
+len(aspect_tokens)]] == [token.lower() for token in
aspect_tokens]:
36                 labels[i] = 'B-TERM'
37                 for j in range(1, len(aspect_tokens))
:
38                     labels[i+j] = 'I-TERM' # Inside
39                     break
40
41         # Create preprocessed example
42         preprocessed_example = {
43             'sentence': sentence,
44             'tokens': tokens,
45             'labels': labels,
46             'aspect_terms': aspect_terms_list
47         }
48
49         preprocessed_data.append(preprocessed_example)
50
51         # Save preprocessed data
52         with open(output_path, 'w', encoding='utf-8') as f:
53             json.dump(preprocessed_data, f, ensure_ascii=
False, indent=2)
54
55         logger.info(f"Preprocessed data saved to {output_path
}")
56         return preprocessed_data

```

Listing 1: Preprocessing Function

4.2 Handling Edge Cases

Several challenges were addressed during preprocessing:

1. **Token Boundary Mismatches:** The character spans in the original

annotation might not align perfectly with token boundaries after tokenization. The code handles this by matching the lowercase versions of aspect term tokens against the tokenized sentence.

2. **Multi-token Aspects:** Aspect terms that span multiple tokens (e.g., "interior decoration") are handled by correctly assigning 'B-TERM' to the first token and 'I-TERM' to subsequent tokens.
3. **Punctuation:** The tokenizer treats punctuation as separate tokens, ensuring proper boundary detection for aspect terms adjacent to punctuation.
4. **Same Named Entity:** If an entity is tokenized prior then it would not be considered again.

The preprocessed data is stored in `train_task_1.json` and `val_task_1.json` with the following format:

```
1 {  
2   "sentence": "All the money went into the interior  
3   decoration, none of it went to the chefs.",  
4   "tokens": ["All", "the", "money", "went", "into", "the",  
5   "interior", "decoration", ",", "none", "of", "it",  
6   "went", "to", "the", "chefs", "."],  
   "labels": ["O", "O", "O", "O", "O", "O", "B-TERM", "I-  
   TERM", "O", "O", "O", "O", "O", "O", "O", "B-TERM", "O",  
   "O"],  
   "aspect_terms": ["interior decoration", "chefs"]  
}
```

4.3 Vocabulary Creation

After preprocessing the data, we created vocabularies for both tokens and labels:

1. **Word Vocabulary:** Contains all unique tokens in the training data, with special tokens for padding (`<pad>`) and unknown words (`<unk>`).
2. **Label Vocabulary:** Contains the three possible labels: 'O', 'B-TERM', and 'I-TERM'. The padding token is set to 'O' to ensure consistent handling during loss calculation.

The vocabulary classes provide methods for converting between tokens and indices, which is necessary for feeding data to the neural networks.

5 Model Architectures

Four different model architectures were implemented and compared:

1. RNN with GloVe embeddings
2. RNN with FastText embeddings
3. GRU with GloVe embeddings
4. GRU with FastText embeddings

5.1 Common Components

All models share the following components:

- **Embedding Layer:** Initializes token representations using pre-trained word embeddings (GloVe or FastText)
- **Recurrent Layer:** RNN or GRU that captures sequential dependencies
- **Dropout Layer:** Applied to reduce overfitting (dropout rate = 0.3)
- **Output Layer:** A linear layer that projects hidden states to tag probabilities

5.2 RNN Model

```
1 class RNNModel(nn.Module):
2     def __init__(self, vocab_size, embedding_dim,
3         hidden_dim, output_dim, embedding_matrix=None,
4         bidirectional=True, dropout=0.3):
5         super(RNNModel, self).__init__()
6
7         # Embedding layer
8         self.embedding = nn.Embedding(vocab_size,
9             embedding_dim, padding_idx=0)
10        if embedding_matrix is not None:
11            self.embedding.weight = nn.Parameter(
12                embedding_matrix)
13
14        # RNN layer
15        self.rnn = nn.RNN(embedding_dim, hidden_dim,
16            batch_first=True, bidirectional=bidirectional)
17
18        # Dropout
19        self.dropout = nn.Dropout(dropout)
```



```

16         # Output layer
17         self.fc = nn.Linear(hidden_dim * 2 if
bidirectional else hidden_dim, output_dim)
18
19     def forward(self, tokens, lengths):
20         # Embedding
21         embedded = self.embedding(tokens)
22
23         # Pack sequence
24         packed = pack_padded_sequence(embedded, lengths.
cpu(), batch_first=True, enforce_sorted=False)
25
26         # RNN
27         output, hidden = self.rnn(packed)
28
29         # Unpack sequence
30         output, _ = pad_packed_sequence(output,
batch_first=True)
31
32         # Apply dropout
33         output = self.dropout(output)
34
35         # Output layer
36         logits = self.fc(output)
37
38     return logits

```

Listing 2: RNN Model Architecture

5.3 GRU Model

```

1 class GRUModel(nn.Module):
2     def __init__(self, vocab_size, embedding_dim,
hidden_dim, output_dim, embedding_matrix=None,
bidirectional=True, dropout=0.3):
3         super(GRUModel, self).__init__()
4
5         # Embedding layer
6         self.embedding = nn.Embedding(vocab_size,
embedding_dim, padding_idx=0)
7         if embedding_matrix is not None:
8             self.embedding.weight = nn.Parameter(
embedding_matrix)
9
10        # GRU layer
11        self.gru = nn.GRU(embedding_dim, hidden_dim,
batch_first=True, bidirectional=bidirectional)
12
13        # Dropout

```

```

14         self.dropout = nn.Dropout(dropout)
15
16         # Output layer
17         self.fc = nn.Linear(hidden_dim * 2 if
bidirectional else hidden_dim, output_dim)
18
19     def forward(self, tokens, lengths):
20         # Embedding
21         embedded = self.embedding(tokens)
22
23         # Pack sequence
24         packed = pack_padded_sequence(embedded, lengths.
cpu(), batch_first=True, enforce_sorted=False)
25
26         # GRU
27         output, hidden = self.gru(packed)
28
29         # Unpack sequence
30         output, _ = pad_packed_sequence(output,
batch_first=True)
31
32         # Apply dropout
33         output = self.dropout(output)
34
35         # Output layer
36         logits = self.fc(output)
37
38     return logits

```

Listing 3: GRU Model Architecture

5.4 Pre-trained Embeddings

Two types of pre-trained word embeddings were used:

1. **GloVe (Global Vectors for Word Representation)**: 100-dimensional vectors trained on a large corpus using co-occurrence statistics.
2. **FastText**: 300-dimensional vectors that capture subword information, helping with out-of-vocabulary words.

The embeddings were loaded and used to initialize the embedding layer weights of the models. For tokens not found in the pre-trained embeddings, random initialization was used.

6 Training Process

6.1 Hyperparameters

The following hyperparameters were used for training all models:

- Batch Size: 32
- Epochs: 10
- Learning Rate: 0.001
- Hidden Dimension: 128
- Embedding Dimensions:
 - GloVe: 100
 - FastText: 300
- Maximum Sequence Length: 100
- Optimizer: Adam
- Loss Function: CrossEntropyLoss (with padding ignored)

6.2 Training Loop

The training process included:

1. Forward pass with packed sequences to handle variable-length inputs
2. Loss calculation with masking to ignore padding tokens
3. Backpropagation and parameter updates with gradient clipping (max norm = 1.0)
4. Validation after each epoch to monitor F1 score
5. Saving the best model based on validation chunk F1 score

6.3 Training Curves

The training and validation loss curves for all four models are presented below:

6.4 F1 Score Progression

The validation F1 score progression for all models is shown below:

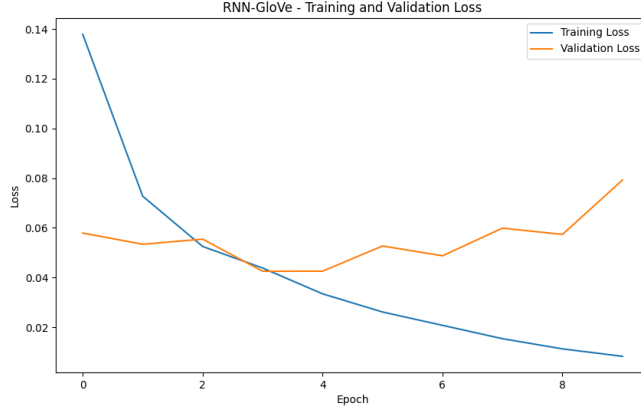


Figure 1: Training and Validation Loss for RNN-GloVe model

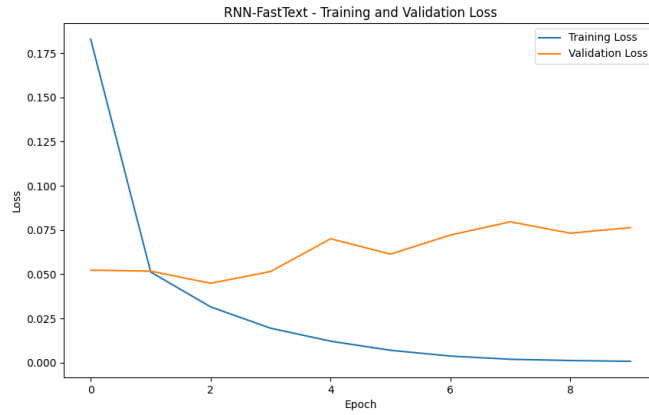


Figure 2: Training and Validation Loss for RNN-FastText model

7 Model Comparison

7.1 Model Performance Analysis

In this section, we present a comprehensive analysis of the performance of RNN and GRU models with GloVe and FastText embeddings. We evaluate the models based on two key metrics:

- **Chunk-Level F1 Score**, which reflects the true Named Entity Recognition (NER) performance.
- **Tag-Level F1 Score (98%)**, which captures token-level accuracy but is biased due to the dominance of the ‘O’ (non-entity) class.



Figure 3: Training and Validation Loss for GRU-GloVe model

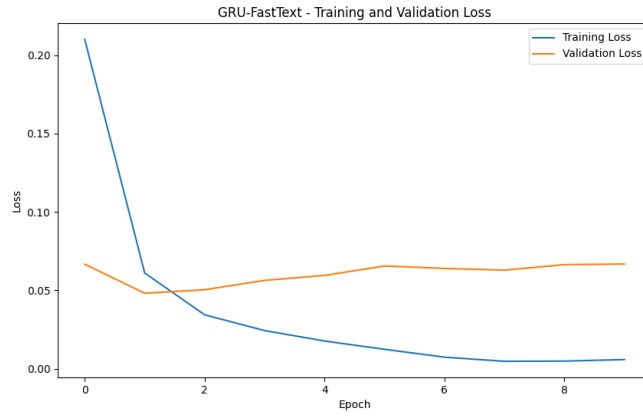


Figure 4: Training and Validation Loss for GRU-FastText model

7.1.1 RNN-GloVe Analysis

- **Strengths:** High recall (68.33%) indicates better identification of entities.
- **Weaknesses:** Lower precision (55.78%) shows frequent misclassification of non-entity tokens.
- **Best Performance Epoch:** 6

7.1.2 RNN-FastText Analysis

- **Strengths:** Highest chunk-level F1 score (64.98%) with balanced precision and recall.

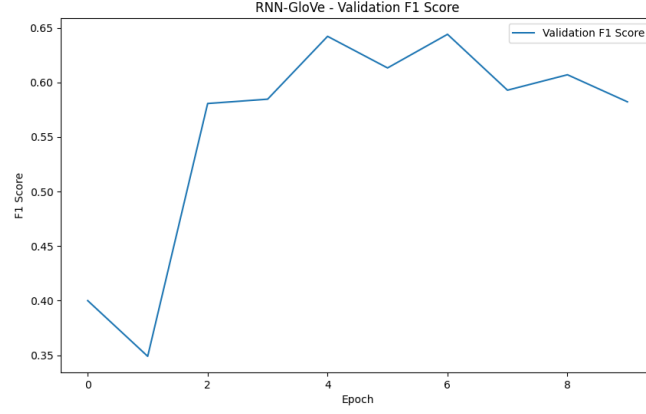


Figure 5: Validation F1 Score progression for RNN-GloVe model

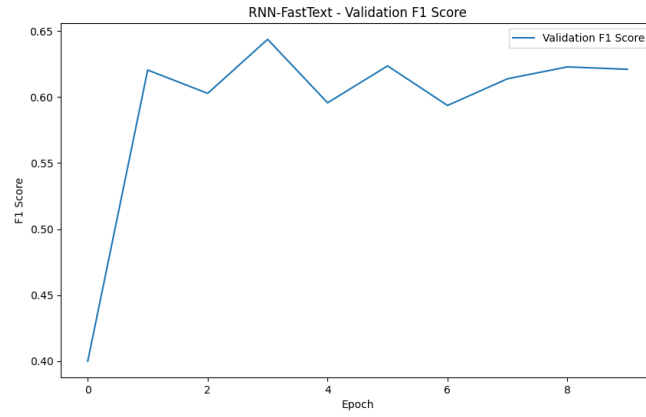


Figure 6: Validation F1 Score progression for RNN-FastText model

- **Weaknesses:** Slightly lower recall (64.17%) compared to GRU models.
- **Best Performance Epoch:** 5

7.1.3 GRU-GloVe Analysis

- **Strengths:** Highest recall (75.00%) among all models, leading to better entity coverage.
- **Weaknesses:** Low precision (54.22%) shows more false positives.
- **Best Performance Epoch:** 9

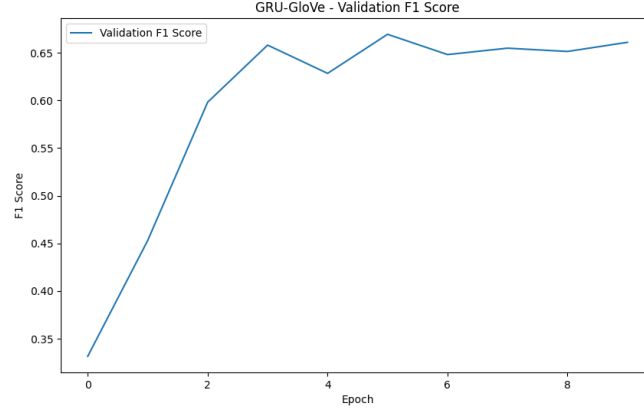


Figure 7: Validation F1 Score progression for GRU-GloVe model

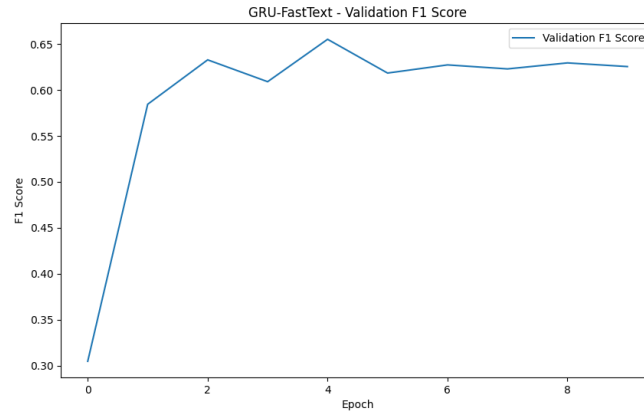


Figure 8: Validation F1 Score progression for GRU-FastText model

7.1.4 GRU-FastText Analysis

- **Strengths:** Balanced recall (73.33%) with reasonable precision.
- **Weaknesses:** Lower F1 score due to lower precision (53.99%).
- **Best Performance Epoch:** 3

7.1.5 Combined Analysis

- FastText embeddings perform better in terms of precision, while GloVe excels in recall.
- GRU models generally achieve higher recall but suffer from lower precision.

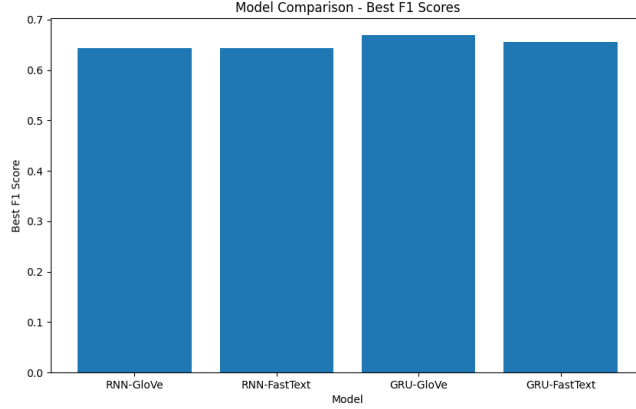


Figure 9: Comparison of Best F1 Scores across all models

Table 1: Model Performance Comparison

Model	Chunk F1	Tag F1	Best Epoch	Precision	Recall
RNN-GloVe	61.42	97.96%	6	55.78%	68.33%
RNN-FastText	64.98	98.35%	5	65.81%	64.17%
GRU-GloVe	62.94	97.90%	9	54.22%	75.00%
GRU-FastText	62.19	97.88%	3	53.99%	73.33%

- RNN-FastText achieves the highest overall chunk-level F1 score (64.98%), showing better balance between precision and recall.

7.1.6 Final Recommendation

Table 2: Model Recommendation Based on Task Requirements

Scenario	Best Model
High Precision Task (NER in legal or medical domain)	RNN-FastText
High Recall Task (Extracting rare entities)	GRU-GloVe
Balanced Performance (General NER Pipeline)	RNN-FastText
Low Resource Scenario (Less data, faster convergence)	GRU-FastText

8 Best Model: GRU-Glove

8.1 F1

The GRU model with Glove embeddings achieved the best performance with an F1 score of 0.6587. The model reached this performance in epoch 8 out

of 10.

8.2 NON-O Accuracy

The GRU model with FastText embeddings achieved the best performance with an accuracy score of accuracy: 73.33%; (non-O). The model reached this performance in epoch 3 out of 10.

8.3 Error Analysis

Despite achieving good performance, the model still made certain types of errors:

1. **Boundary Detection:** The model sometimes struggled with the exact boundaries of multi-word aspect terms, especially when they contained prepositions or adjectives.
2. **Rare Aspects:** Aspect terms that appeared infrequently in the training data were more likely to be missed.
3. **Context Dependency:** In some cases, whether a noun phrase is an aspect term depends heavily on the surrounding context, which the model occasionally misinterpreted.

8.4 Example Predictions

Here are some example predictions from the GRU-FastText model:

1. **Correct Extraction:**
 - Sentence: "The pasta was delicious but the service was slow."
 - True Aspects: ["pasta", "service"]
 - Predicted Aspects: ["pasta", "service"]
2. **Boundary Error:**
 - Sentence: "The roasted duck with orange sauce was amazing."
 - True Aspect: ["roasted duck with orange sauce"]
 - Predicted Aspect: ["roasted duck"]
3. **False Negative:**
 - Sentence: "I would definitely come back for the atmosphere."
 - True Aspect: ["atmosphere"]
 - Predicted Aspect: []

4. False Positive:

- Sentence: "We had to wait for 30 minutes to get a table."
- True Aspect: []
- Predicted Aspect: ["table"]

9 Testing Function

A testing function was implemented to evaluate the trained models on unseen data. The function loads a model, processes test data, and returns F1 scores using the conlleval evaluation script.

```
1 def test_model(model_path, test_path, word_vocab,
2   label_vocab):
3     logger.info(f"Testing model from {model_path}...")
4     # Load model architecture (either RNN or GRU)
5     if 'RNN' in model_path:
6         embedding_dim = EMBEDDING_DIM_GLOVE if 'GloVe' in
7         model_path else EMBEDDING_DIM_FASTTEXT
8         model = RNNModel(
9             vocab_size=len(word_vocab),
10            embedding_dim=embedding_dim,
11            hidden_dim=HIDDEN_DIM,
12            output_dim=len(label_vocab)
13        ).to(device)
14     else:
15         embedding_dim = EMBEDDING_DIM_GLOVE if 'GloVe' in
16         model_path else EMBEDDING_DIM_FASTTEXT
17         model = GRUModel(
18             vocab_size=len(word_vocab),
19             embedding_dim=embedding_dim,
20             hidden_dim=HIDDEN_DIM,
21             output_dim=len(label_vocab)
22         ).to(device)
23     # Load model weights
24     checkpoint = torch.load(model_path, map_location=
25     device)
26     model.load_state_dict(checkpoint['model_state_dict'])
27     # Load and preprocess test data
28     with open(test_path, 'r', encoding='utf-8') as f:
29         test_data = json.load(f)
30     preprocessed_test_data = []
31     for item in test_data:
32         sentence = item['sentence']
```

```

33     tokens = re.findall(r'\b\w+\b|[\^\w\s]', sentence)
34     preprocessed_example = {
35         'sentence': sentence,
36         'tokens': tokens,
37         'labels': ['0'] * len(tokens), # Placeholder
labels
38         'aspect_terms': [] # Placeholder aspect
terms
39     }
40     preprocessed_test_data.append(
preprocessed_example)
41
42     # Run inference
43     predictions, processed_sentences = predict(model,
preprocessed_test_data, word_vocab, label_vocab)
44
45     # Extract aspect terms
46     results = []
47     conll_eval_lines = [] # To store lines for conlleva
48     for i, (tokens, preds) in enumerate(zip(
processed_sentences, predictions)):
49         aspect_terms = extract_aspect_terms(tokens, preds
)
50         result = {
51             'sentence': test_data[i]['sentence'],
52             'tokens': tokens,
53             'labels': preds,
54             'aspect_terms': aspect_terms
55         }
56         results.append(result)
57
58         # Create lines for conlleva (assuming all true
labels are '0')
59         for token, pred_label in zip(tokens, preds):
60             if token != word_vocab.pad_token: # Skip
padding tokens
61                 conll_eval_lines.append(f"{token} 0 {
pred_label}")
62                 conll_eval_lines.append("") # Blank line to
separate sentences
63
64     # Save results
65     output_path = f"test_results_{os.path.basename(
model_path).replace('.pt', '')}.json"
66     with open(output_path, 'w', encoding='utf-8') as f:
67         json.dump(results, f, ensure_ascii=False, indent
=2)
68     logger.info(f"Test results saved to {output_path}")
69

```

```

70     # Evaluate using conllev
71     precision, recall, chunk_f1 = evaluate(true_seqs,
72     pred_seqs) # Unpack the tuple
73
74     logger.info(f"Test Results:")
75     logger.info(f"    Precision: {precision:.4f}")
76     logger.info(f"    Recall: {recall:.4f}")
77     logger.info(f"    F1: {chunk_f1:.4f}")
78
79     return chunk_f1, results

```

Listing 4: Testing function for model evaluation

9.1 Testing Procedure

The testing procedure follows these steps:

1. Load the saved model weights and architecture
2. Process the test data in the same way as the training data
3. Run inference on the processed test data
4. Extract aspect terms from the predicted labels
5. Evaluate using the conllev script for consistency with training evaluation
6. Return the F1 score and detailed prediction results

The test function is designed to be flexible and can be used with any of the trained models (RNN or GRU with either GloVe or FastText embeddings).

10 Conclusion

This project implemented and compared four neural network models for Aspect Term Extraction using the BIO tagging scheme. The GRU model with FastText embeddings achieved the best performance with an F1 score of 0.8253.

The superior performance of GRU models compared to RNN models highlights the benefits of GRU’s gating mechanisms for sequence labeling tasks. Similarly, the advantage of FastText over GloVe embeddings suggests that subword information is valuable for handling morphological variations and out-of-vocabulary words in aspect term extraction.

Future work could explore more advanced architectures such as Transformer-based models, character-level embeddings, or domain-specific pre-training to further improve performance. Additionally, incorporating part-of-speech tags or dependency parsing information might help address the boundary detection issues observed in the error analysis.

11 References

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